SALES DATA ANALYSIS DASHBOARD - Excel

Microsoft Excel | Retail Sales Performance

Tool	Microsoft Excel (Office 365)
Dataset	Retail Sales Transactions
Time Period	2014 – 2017
Dataset Size	~9,994 Rows × 11 Columns
Deliverable	Interactive KPI Dashboard + Cleaned Data Model

Project Summary

This project delivers an end-to-end retail sales analytics solution built entirely in Microsoft Excel. Raw transactional data spanning four fiscal years (2014–2017) was ingested, cleaned, and modelled using Power Query and Excel formulas, then surfaced through an interactive, slicer-driven KPI dashboard designed for business stakeholders.

Dimension	Detail
Goal	Identify revenue drivers, profit patterns, and growth opportunities across product categories, geographies, and customer segments
Data Source	Retail order transaction log — 9,994 rows × 11 columns (Order Date, Year, Month, Customer, State, Category, Sub-Category, Product, Sales, Quantity, Profit)
What Was Built	4-sheet workbook: Raw Data → Cleaned Data → Pivot Layer → Executive Dashboard
Key KPIs	Total Sales, Total Profit, Profit Margin %, YoY Growth %, Customer Count, Top 5 Profit Customers
Outcome	Technology and Office Supplies drove highest profit growth; Q4 accounts for ~35% of annual revenue; top 5 customers contribute a disproportionate share of profits

Problem Statement

The retail business lacked a centralised, self-service view of sales performance. Decision-makers had to manually sift through raw transaction exports to understand which products, regions, and customers were driving — or eroding — profitability. This project addresses the following business questions:

- Which product sub-categories generate the highest revenue and the strongest profit margins?
- How have sales and profits trended across the four-year period (2014–2017)?
- Which US states are the top contributors to total revenue?
- Is the customer base growing year-over-year, and at what rate?
- Which months exhibit peak and trough sales patterns, and why?
- Who are the top 5 customers by profit contribution, and how concentrated is revenue risk?

Objectives

- Clean and standardise raw transaction data (handle nulls, date formats, duplicate records, currency inconsistencies)
- Build a structured Pivot layer to aggregate Sales, Profit, and Quantity across Category, Sub-Category, State, Month, and Year dimensions
- Calculate KPIs — Total Sales, Total Profit, Profit Margin %, YoY Growth %, Customer Count
- Design an executive-grade interactive dashboard with Category and Year slicers for real-time data filtering
- Visualise sub-category performance (horizontal bar), profit trends (multi-line chart), state sales (choropleth map), customer count by year (bar), monthly sales (area chart), and top-5 customer profits (pie chart)
- Enable non-technical stakeholders to extract insights without SQL or BI tool knowledge

Data & Cleaning

Cleaning Steps Performed

- Removed 11 duplicate order records identified via conditional formatting + COUNTIFS
- Standardised Order Date to YYYY-MM-DD format using Power Query date parsing
- Derived Year and Month columns using YEAR() / MONTH() functions for pivot aggregation
- Formatted Sales and Profit columns as Accounting (USD, 2 decimal places)
- Verified no blank cells in key columns (Customer Name, State, Category) using COUNTA checks
- Trimmed whitespace from Category and Sub-Category text fields using TRIM() + CLEAN()
- Cross-validated row count before/after cleaning: 9,994 → 9,983 clean records

Excel Skills & Features Used

Feature / Technique	Applied For	Status
Pivot Tables	Aggregating Sales, Profit, Quantity by Category, State, Month, Year	✓ Used
Pivot Charts	Bar, Line, Area, Pie, Map charts sourced from Pivot Tables	✓ Used
Slicers	Interactive filtering by Category (3 values) and Year (2014–2017)	✓ Used
Power Query	Data import, type casting, date parsing, whitespace trimming	✓ Used
SUMIFS / COUNTIFS	Conditional aggregation for KPI cards and validation checks	✓ Used
XLOOKUP	Customer profit lookup for dynamic Top-5 table	✓ Used
Conditional Formatting	Heat-map shading on Sub-Category bar chart and KPI delta indicators	✓ Used
Data Validation	Dropdown lists to ensure consistent Category / Year filter inputs	✓ Used
Named Ranges	Clean, readable formula references across sheets	✓ Used
Charts (Bar / Line / Pie / Map / Area)	6 distinct chart types on the dashboard canvas	✓ Used
Dashboard Layout & Formatting	Rounded-corner card borders, icon headers, unified blue theme	✓ Used
YEAR() / MONTH() Functions	Date dimension extraction for time-series pivots	✓ Used

KPI Definitions

KPI	Formula / Definition	Unit
Total Sales	SUM(Sales)	USD $
Total Profit	SUM(Profit)	USD $
Profit Margin %	Total Profit / Total Sales × 100	% (e.g., 12.47%)
YoY Revenue Growth %	(Current Year Sales − Prior Year Sales) / Prior Year Sales × 100	%
Customer Count	COUNTA(UNIQUE(Customer Name)) per Year	Count
Top 5 Customer Profit	LARGE(Profit by Customer, 1–5) via Pivot + XLOOKUP	USD $
Avg. Order Value	Total Sales / COUNT(Orders)	USD $

Key Insights

1. **Phones & Chairs are the top two sub-categories by revenue** — $330,007 and $328,168 respectively, together ~18% of total sales
2. **Technology is the highest-profit category, with accelerating growth** — profit grew from ~$20K (2014) to ~$50K+ (2017), a ~150% increase over 4 years
3. **Office Supplies shows the most consistent profit trajectory** — steady upward trend 2014–2017 with lower variance than Furniture
4. **Furniture has the weakest profit profile despite strong sales** — high sales volume but low/negative margin on sub-categories like Tables and Bookcases
5. **Q4 (Oct–Dec) is the dominant sales period** — monthly sales chart shows a pronounced spike in months 9–12, accounting for ~35% of annual revenue
6. **Customer base grew consistently YoY from 2014 to 2017** — customer count bars show ~15–20% YoY increase in unique buyers
7. **Top 5 customers disproportionately drive profits** — Tamara Chand, Raymond Buch, Sanjit Chand, Hunter Lopez, and Adrian Barton generate a large share of total profit, creating concentration risk
8. **A small number of states dominate sales** — choropleth map shows $457,688 as the highest state-level sales figure, with California leading significantly

Conclusion

Value Delivered

- Replaced manual CSV-level analysis with a single-pane, interactive Excel dashboard accessible to non-technical stakeholders
- Surfaced actionable insights on category profitability, regional concentration, customer risk, and seasonal patterns
- Demonstrated end-to-end Excel data engineering: raw data → cleaning → pivot modelling → visualisation

Dashboard Image :